

CS372 Lab1: HTTP

1. The Basic HTTP GET/response interaction

1. Is your browser running HTTP version 1.0 or 1.1? What version of HTTP is the server running?

My browser runs the HTTP 1.1 version, and the same version of HTTP is the server running.

No.	Time	Source	Destination	Protocol	Length	Info
392	23:20:40.8680...	192.168.68.62	128.119.245.12	HTTP	532	GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1
406	23:20:41.0179...	128.119.245.12	192.168.68.62	HTTP	540	HTTP/1.1 200 OK (text/html)
415	23:20:41.1985...	192.168.68.62	128.119.245.12	HTTP	478	GET /favicon.ico HTTP/1.1
434	23:20:41.3183...	128.119.245.12	192.168.68.62	HTTP	538	HTTP/1.1 404 Not Found (text/html)

2. What languages (if any) does your browser indicate that it can accept to the Server? English (US)

The image shows a Wireshark packet capture of an HTTP GET request. The packet list on the left shows four packets. The selected packet (No. 392) is expanded in the packet details pane on the right. The details pane shows the following information:

- Ethernet II, Src: 2e:bb:2a:fc:3d:c3 (2e:bb:2a:fc:3d:c3), Dst: 128.119.245.12
- Internet Protocol Version 4, Src: 192.168.68.62, Dst: 128.119.245.12
- Transmission Control Protocol, Src Port: 53744, Dst Port: 80
- Hypertext Transfer Protocol
 - GET /wireshark-labs/HTTP-wireshark-file1.html HTTP/1.1\r\n
 - Request Method: GET
 - Request URI: /wireshark-labs/HTTP-wireshark-file1.html
 - Request Version: HTTP/1.1
 - Host: gaia.cs.umass.edu\r\n
 - Connection: keep-alive\r\n
 - Upgrade-Insecure-Requests: 1\r\n
 - User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7; rv:68.0) Gecko/20100101 Firefox/68.0\r\n
 - Accept: text/html,application/xhtml+xml,application/xml;q=0.9,*/*;q=0.8\r\n
 - Accept-Encoding: gzip, deflate\r\n
 - Accept-Language: en-US,en;q=0.9\r\n

3. What is the IP address of your computer? Of the gaia.cs.umass.edu server?

My Computer IP address: 192.168.68.62, UMASS IP address: 128.119.245.12

> Internet Protocol Version 4, Src: 192.168.68.62, Dst: 128.119.245.12
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4. What is the status code returned from the server to your browser?

The image shows a Wireshark packet capture of an HTTP 200 OK response. The packet list on the left shows four packets. The selected packet (No. 406) is expanded in the packet details pane on the right. The details pane shows the following information:

- Hypertext Transfer Protocol
 - HTTP/1.1 200 OK\r\n
 - Response Version: HTTP/1.1
 - Status Code: 200
 - [Status Code Description: OK]
 - Response Phrase: OK
 - Date: Mon, 27 Jan 2025 07:20:41 GMT\r\n
 - Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips\r\n
 - Last-Modified: Mon, 27 Jan 2025 06:59:01 GMT\r\n
 - Etag: "80-62caa9c7a501d"\r\n

200 Status Code

5. When was the HTML file that you are retrieving last modified at the server?

Last-Modified: Mon, 27 Jan 2025 06:59:01 GMT

```
Last-Modified: Mon, 27 Jan 2025 06:59:01 GMT\r\n
```

6. How many bytes of content are being returned to your browser?

128 bytes

```
[Request in frame: 392]
[Time since request: 0.149840000 seconds]
[Request URI: /wireshark-labs/HTTP-wireshark-file2.html]
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]
File Data: 128 bytes
```

7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name One.

No, All data appears to be displayed in the packet-listing window.

2. The HTTP CONDITIONAL GET/response interaction

8. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

No, I didn't see the line in the HTTP GET:

The image shows a Wireshark packet capture of an HTTP interaction. The packet list at the top shows four packets: a GET request (No. 95), a 200 OK response (No. 113), another GET request (No. 248), and a 304 Not Modified response (No. 249). The packet details pane for packet 95 is expanded, showing the Hypertext Transfer Protocol section. A red box is drawn around the GET request details, and a large red 'NO' is written to the left of it, indicating that the 'IF-MODIFIED-SINCE' header is not present in the request.

```
Frame 95: 532 bytes on wire (4256 bits), 532 bytes captured (4256 bits) on interface en0, id 0
  Ethernet II, Src: 2e:bb:2a:fc:3d:c3 (2e:bb:2a:fc:3d:c3), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac)
  Internet Protocol Version 4, Src: 192.168.68.62, Dst: 128.119.245.12
  Transmission Control Protocol, Src Port: 58332, Dst Port: 80, Seq: 1, Ack: 1, Len: 478
  Hypertext Transfer Protocol
    GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Upgrade-Insecure-Requests: 1\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko)
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
    [Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]
```

9. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

Yes, the server explicitly returns the contents of the file, I can tell from the line-based text data, which contains the HTML text.

```
File Data: 371 bytes
Line-based text data: text/html (10 lines)
\n
<html>\n
\n
Congratulations again! Now you've downloaded the file lab2-2.html. <br>\n
This file's last modification date will not change. <p>\n
Thus if you download this multiple times on your browser, a complete copy <br>\n
will only be sent once by the server due to the inclusion of the IN-MODIFIED-SINCE<br>\n
field in your browser's HTTP GET request to the server.\n
\n
</html>\n
```

10. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

Yes, I saw the line in the second HTTP GET, it was followed by a time, Mon,27, Jan 2025 06:59:01 GMT

The image shows a Wireshark packet capture of an HTTP transaction. The packet list at the top shows four packets: a GET request (frame 95), an HTTP 200 OK response (frame 113), another GET request (frame 248), and an HTTP 304 Not Modified response (frame 249). The packet details pane for frame 248 is expanded, showing the request headers. The 'If-Modified-Since' header is highlighted with a red box, showing the value 'Mon, 27 Jan 2025 06:59:01 GMT'. The status bar at the bottom indicates the response is in frame 249 and provides the full request URI.

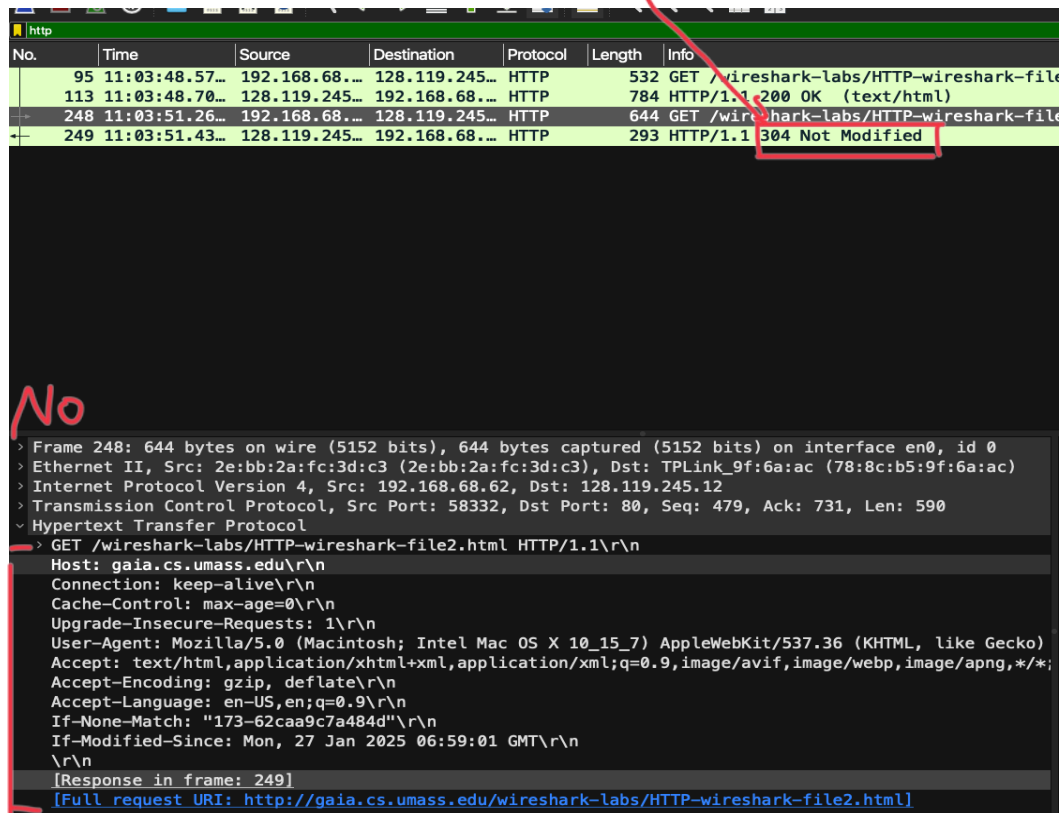
No.	Time	Source	Destination	Protocol	Length	Info
95	11:03:48.57...	192.168.68...	128.119.245...	HTTP	532	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
113	11:03:48.70...	128.119.245...	192.168.68...	HTTP	784	HTTP/1.1 200 OK (text/html)
248	11:03:51.26...	192.168.68...	128.119.245...	HTTP	644	GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1
249	11:03:51.43...	128.119.245...	192.168.68...	HTTP	293	HTTP/1.1 304 Not Modified

Frame 248: 644 bytes on wire (5152 bits), 644 bytes captured (5152 bits) on interface en0, id 0
Ethernet II, Src: 2e:bb:2a:fc:3d:c3 (2e:bb:2a:fc:3d:c3), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac)
Internet Protocol Version 4, Src: 192.168.68.62, Dst: 128.119.245.12
Transmission Control Protocol, Src Port: 58332, Dst Port: 80, Seq: 479, Ack: 731, Len: 590
Hypertext Transfer Protocol
GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\nHost: gaia.cs.umass.edu\r\nConnection: keep-alive\r\nCache-Control: max-age=0\r\nUpgrade-Insecure-Requests: 1\r\nUser-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*; Accept-Encoding: gzip, deflate\r\nAccept-Language: en-US,en;q=0.9\r\nIf-None-Match: "1122-62aa0c7a484d"\r\nIf-Modified-Since: Mon, 27 Jan 2025 06:59:01 GMT\r\n\r\n[Response in frame: 249]
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

11. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

The second HTTP status code is: 304 Not Modified.

The server didn't respond with the contents of the file because the browser loaded it from its cache



No.	Time	Source	Destination	Protocol	Length	Info
95	11:03:48.57...	192.168.68...	128.119.245...	HTTP	532	GET /wireshark-labs/HTTP-wireshark-file...
113	11:03:48.70...	128.119.245...	192.168.68...	HTTP	784	HTTP/1.1 200 OK (text/html)
248	11:03:51.26...	192.168.68...	128.119.245...	HTTP	644	GET /wireshark-labs/HTTP-wireshark-file...
249	11:03:51.43...	128.119.245...	192.168.68...	HTTP	293	HTTP/1.1 304 Not Modified

No

> Frame 248: 644 bytes on wire (5152 bits), 644 bytes captured (5152 bits) on interface en0, id 0
> Ethernet II, Src: 2e:bb:2a:fc:3d:c3 (2e:bb:2a:fc:3d:c3), Dst: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac)
> Internet Protocol Version 4, Src: 192.168.68.62, Dst: 128.119.245.12
> Transmission Control Protocol, Src Port: 58332, Dst Port: 80, Seq: 479, Ack: 731, Len: 590
> Hypertext Transfer Protocol
GET /wireshark-labs/HTTP-wireshark-file2.html HTTP/1.1\r\nHost: gaia.cs.umass.edu\r\nConnection: keep-alive\r\nCache-Control: max-age=0\r\nUpgrade-Insecure-Requests: 1\r\nUser-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*; Accept-Encoding: gzip, deflate\r\nAccept-Language: en-US,en;q=0.9\r\nIf-None-Match: "173-62caa9c7a484d"\r\nIf-Modified-Since: Mon, 27 Jan 2025 06:59:01 GMT\r\n\r\n[Response in frame: 249]
[Full request URI: http://gaia.cs.umass.edu/wireshark-labs/HTTP-wireshark-file2.html]

3. Retrieving Long Documents

12. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill of Rights?

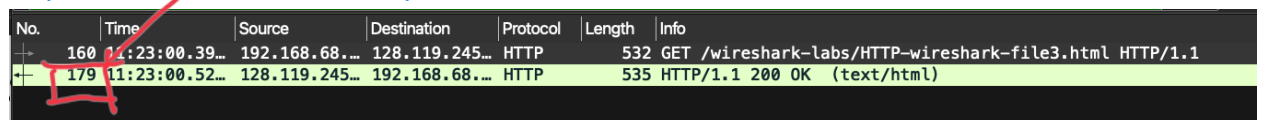
Only one HTTP GET message did my browser send, packet number: 160 in the trace contains the GET message for the Bill of Rights.



No.	Time	Source	Destination	Protocol	Length	Info
160	11:23:00.39...	192.168.68...	128.119.245...	HTTP	532	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1

13. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

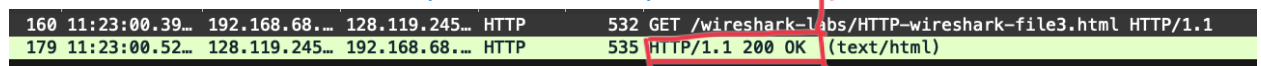
Packet number: 179 in the trace contains the status code and the phrase with the response to the HTTP GET request.



No.	Time	Source	Destination	Protocol	Length	Info
160	11:23:00.39...	192.168.68...	128.119.245...	HTTP	532	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
179	11:23:00.52...	128.119.245...	192.168.68...	HTTP	535	HTTP/1.1 200 OK (text/html)

14. What is the status code and phrase in the response?

The Status code is 200, and the phrase in the response is OK



No.	Time	Source	Destination	Protocol	Length	Info
160	11:23:00.39...	192.168.68...	128.119.245...	HTTP	532	GET /wireshark-labs/HTTP-wireshark-file3.html HTTP/1.1
179	11:23:00.52...	128.119.245...	192.168.68...	HTTP	535	HTTP/1.1 200 OK (text/html)

15. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?

4 TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights.

```
TCP segment data (481 bytes)
[4 Reassembled TCP Segments (4861 bytes): #176(1460), #177(1460), #178(1460), #179(481)]
[Frame: 176, payload: 0-1459 (1460 bytes)]
[Frame: 177, payload: 1460-2919 (1460 bytes)]
[Frame: 178, payload: 2920-4379 (1460 bytes)]
[Frame: 179, payload: 4380-4860 (481 bytes)]
[Segment count: 4]
[Reassembled TCP length: 4861]
```

4. HTML Documents with Embedded Objects

16. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

my browser sent 3 HTTP GET request messages; the first and the second internet address is 128.119.245.12, and the last internet address is 178.79.137.164.

No.	Time	Source	Destination	Protocol	Length	Info
1213	16:40:57.162611	192.168.68.64	178.79.137.164	HTTP	457	GET /8E_cover_small.jpg HTTP/1.1
1072	16:40:56.854773	192.168.68.64	128.119.245.12	HTTP	490	GET /pearson.png HTTP/1.1
1028	16:40:56.744207	192.168.68.64	128.119.245.12	HTTP	544	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
1083	16:40:56.945239	128.119.245.12	192.168.68.64	HTTP	781	HTTP/1.1 200 OK (PNG)
1064	16:40:56.842009	128.119.245.12	192.168.68.64	HTTP	1367	HTTP/1.1 200 OK (text/html)
1217	16:40:57.336258	178.79.137.164	192.168.68.64	HTTP	237	HTTP/1.1 301 Moved Permanently

17. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two websites in parallel? Explain.

From the listing packet, my browser downloaded the two images serially, because the GET request for /pearson.png happened at 16:40:56.854773 and the request for /8E_cover_small.jpg followed at 16:40:57.162611, there is indeed a slight delay between the two requests. This indicates that the images were downloaded serially, not in parallel.

No.	Time	Source	Destination	Protocol	Length	Info
328	12:14:59.007331	192.168.68.62	128.119.245.12	HTTP	532	GET /wireshark-labs/HTTP-wireshark-file4.html HTTP/1.1
344	12:14:59.104646	128.119.245.12	192.168.68.62	HTTP	1355	HTTP/1.1 200 OK (text/html)
346	12:14:59.132321	192.168.68.62	128.119.245.12	HTTP	478	GET /pearson.png HTTP/1.1
362	12:14:59.236425	128.119.245.12	192.168.68.62	HTTP	745	HTTP/1.1 200 OK (PNG)
372	12:14:59.429523	192.168.68.62	178.79.137.164	HTTP	457	GET /8E_cover_small.jpg HTTP/1.1
374	12:14:59.820499	178.79.137.164	192.168.68.62	HTTP	237	HTTP/1.1 301 Moved Permanently

Sorry, I re-did this part

5 HTTP Authentication

18. What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?

The server responded with a status code 401 with the phrase Unauthorized.

No.	Time	Source	Destination	Protocol	Length	Info
476	...	192.168.68.62	128.119.245.12	HTTP	548	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
504	...	128.119.245.12	192.168.68.62	HTTP	771	HTTP/1.1 401 Unauthorized (text/html)
705	...	192.168.68.62	128.119.245.12	HTTP	633	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
708	...	128.119.245.12	192.168.68.62	HTTP	544	HTTP/1.1 200 OK (text/html)


```

> Frame 504: 771 bytes on wire (6168 bits), 771 bytes captured (6168 bits) on interface en0, id 0
> Ethernet II, Src: TPLink_9f:6a:ac (78:8c:b5:9f:6a:ac), Dst: 2e:bb:2a:fc:3d:c3 (2e:bb:2a:fc:3d:c3)
> Internet Protocol Version 4, Src: 128.119.245.12, Dst: 192.168.68.62
> Transmission Control Protocol, Src Port: 80, Dst Port: 58976, Seq: 1, Ack: 495, Len: 717
> Hypertext Transfer Protocol
  > HTTP/1.1 401 Unauthorized\r\n
    Date: Mon, 27 Jan 2025 20:42:00 GMT\r\n
    Server: Apache/2.4.6 (CentOS) OpenSSL/1.0.2k-fips PHP/7.4.33 mod_perl/2.0.11 Perl/v5.16.3\r\n
    WWW-Authenticate: Basic realm="wireshark-students only"\r\n
    Content-Length: 381\r\n
    Keep-Alive: timeout=5, max=100\r\n
    Connection: Keep-Alive\r\n
    Content-Type: text/html; charset=iso-8859-1\r\n
    \r\n
    [Request in frame 476]
  
```

19. When your browser's sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?

The new field included in the HTTP GET message is Authorization:

Authorization: Basic d2lyZXNoYXJrLXN0dWRIbnRzOm5ldHdvcms=\r\n

No.	Time	Source	Destination	Protocol	Length	Info
476	...	192.168.68.62	128.119.245.12	HTTP	548	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
504	...	128.119.245.12	192.168.68.62	HTTP	771	HTTP/1.1 401 Unauthorized (text/html)
705	...	192.168.68.62	128.119.245.12	HTTP	633	GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1
708	...	128.119.245.12	192.168.68.62	HTTP	544	HTTP/1.1 200 OK (text/html)


```

> Transmission Control Protocol, Src Port: 58977, Dst Port: 80, Seq: 1, Ack: 1, Len: 579
> Hypertext Transfer Protocol
  > GET /wireshark-labs/protected_pages/HTTP-wireshark-file5.html HTTP/1.1\r\n
    Host: gaia.cs.umass.edu\r\n
    Connection: keep-alive\r\n
    Authorization: Basic d2lyZXNoYXJrLXN0dWRIbnRzOm5ldHdvcms=\r\n
    User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_7) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/
    Accept: text/html,application/xhtml+xml,application/xml;q=0.9,image/avif,image/webp,image/apng,*/*;q=0.8;ap
    Accept-Encoding: gzip, deflate\r\n
    Accept-Language: en-US,en;q=0.9\r\n
    \r\n
  
```